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Waiting for participants to join…



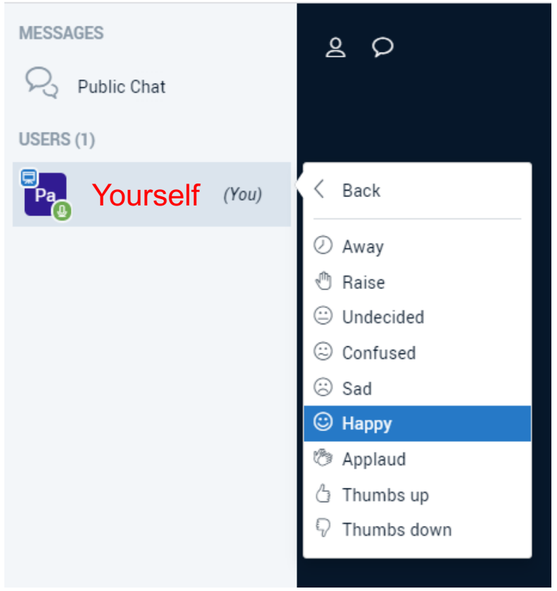
PREPARING A DMP

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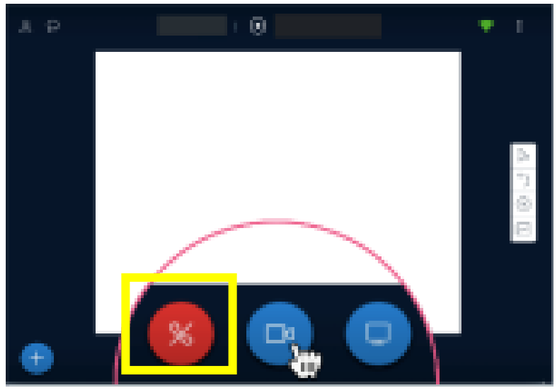
IS EVERYTHING CORRECT?

Can you hear the presenter and moderator?

a. If yes, set your status to ☺



b. If no, click on the telephone icon. It should not be blue



c. If you still have problems, set your status to 👎

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PRE-CHECKS TO AVOID TECHNICAL PROBLEMS

1. **Check compatibility of your system:** leave meeting, perform test, close test window, re-join meeting.
   - https://webrtc.github.io/samples/src/content/devices/input-output/
2. **Use compatible and updated browsers (in order of preference):**
   - Google Chrome, Firefox and Edge
     - Firefox might give problems when sharing screen
3. **Check if your browser has permission to access your devices** (micro, webcam, screen sharing…):
   - https://webrtc.github.io/samples/src/content/devices/input-output/
   - https://www.webrtc-experiment.com/Pluginfree-Screen-Sharing/#726544242382806
4. **Optimize your computer and internet speed**
   - Close other tabs, windows, programs
   - Check your speed (min 2-3 Mbps): https://www.speedtest.net/

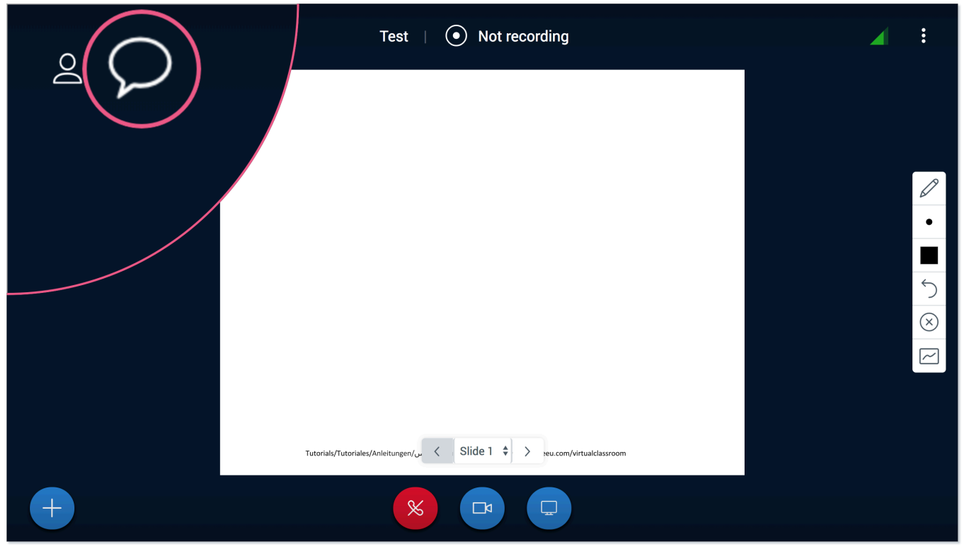
QUICK TROUBLESHOOTING TIPS

1. Reload the page
2. Restart the browser.
   - Do not "Restore all Tabs" when reopening the browser.
3. Open the meeting in a different browser.
4. Open the meeting in an Incognito or Private window.
5. Disconnect from any connected VPNs/wireless hotspots.
6. Leave meeting and re-join.
7. Restart your device.
8. Check email received before the session for more tips.

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HOW THE SESSION WORKS

1. 45’ presentation + 15’ questions
2. Write your questions in the public chat.
   - The moderator will collect them for the end of the session.
3. Participate in polls
4. Session is being recorded



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RESEARCH DEPARTMENT
UNIVERSITY LIBRARY

PREPARING A DATA MANAGEMENT PLAN

RDM webinar series, 05-05-2020

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Slides: https://osf.io/p6jqk

CC BY

LAST WEEK'S POLL

— Did you have to write a Data Management Plan in the past?

A. Yes

B. No, never

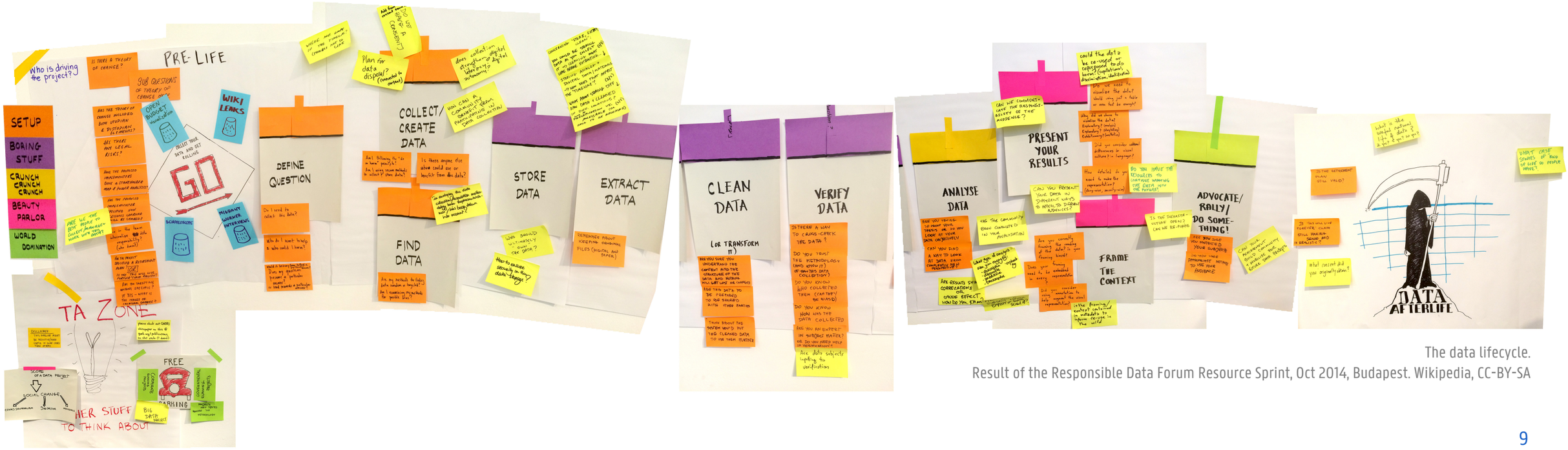
C. No, but I need to submit one for the first time soon



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OUTLINE

- DMPs: what & why?
- External funder & UGent requirements
- What to cover in your (FWO) DMP?
- DMPonline.be
- What other DMP support is available?
- Q&A



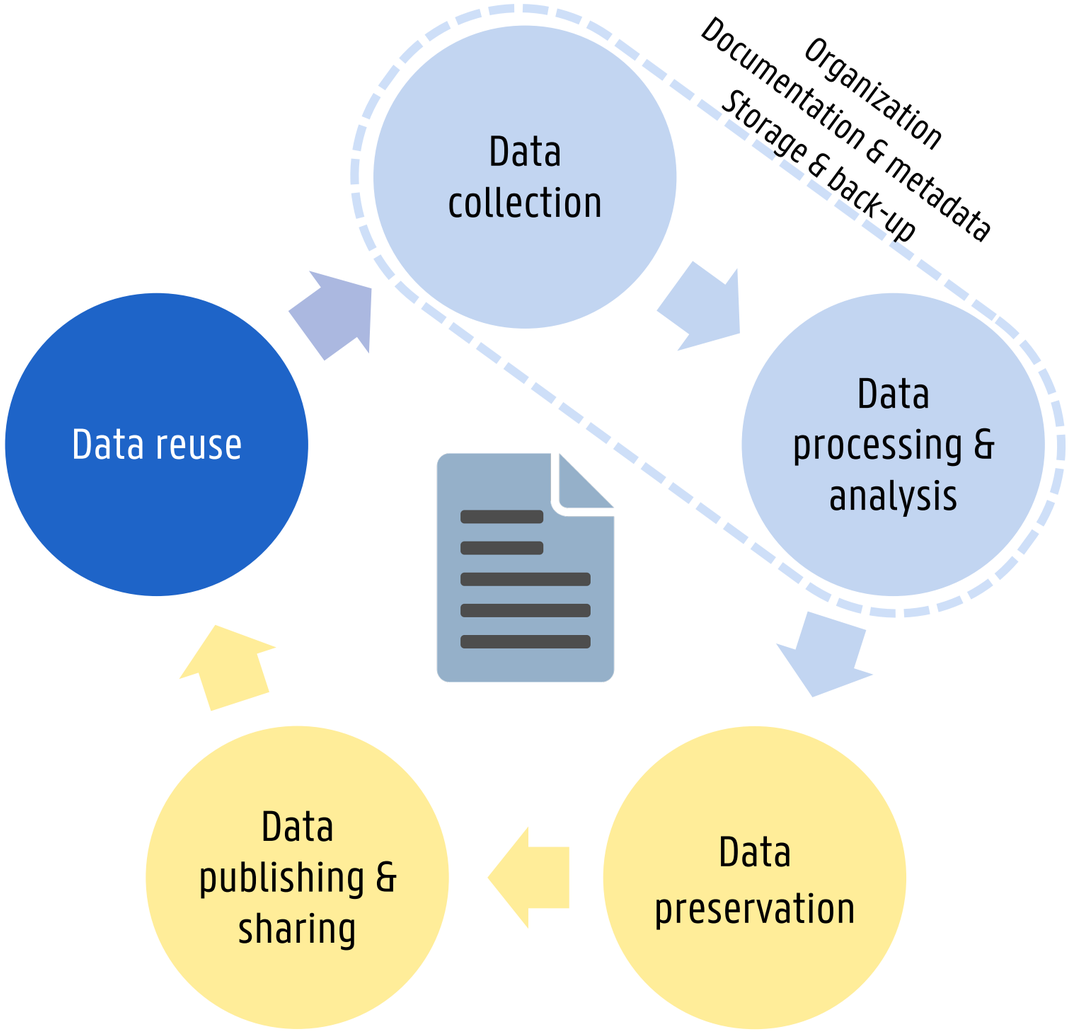
The data lifecycle.
Result of the Responsible Data Forum Resource Sprint, Oct 2014, Budapest. Wikipedia, CC-BY-SA

DMPS: WHAT & WHY?

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DATA MANAGEMENT PLAN (DMP)

- Outlines how research data will be handled during & after a project
- Ideally a 'living' document
- Increasingly required by research funders/institutions as a first step towards good RDM

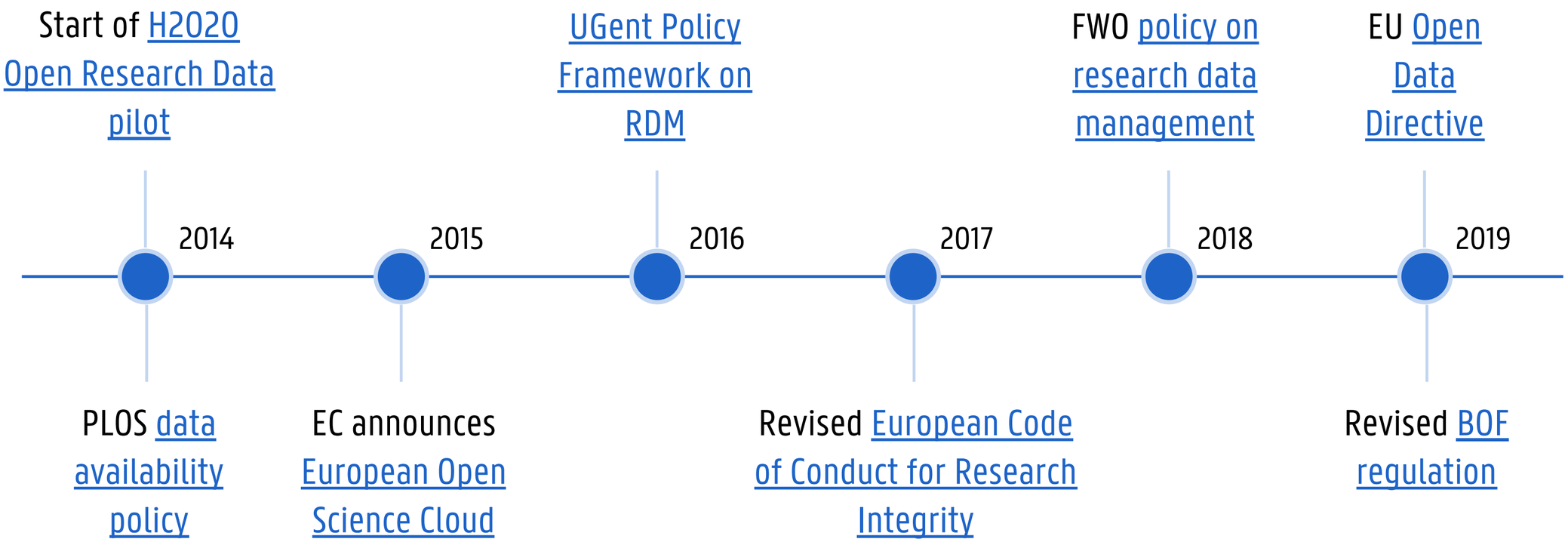


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https://www.ugent.be/en/research/datamanagement/before-research/datamanagementplan.htm

GROWING EXPECTATIONS

Sustainable access to publicly funded research data



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WITHOUT PLANNING…

- Teams/labs often lose (or are unable to find) data when researchers leave
- Working with old data, or data from collaborators, can be problematic
- There often is no budget allocated to maintaining data beyond a project's end
- Gaining consent for archiving and sharing data is often forgotten
- Etc.

→ DMPs aim to help you avoid such headaches

(and assure funders that public money is well spent)

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QUESTION TIME!

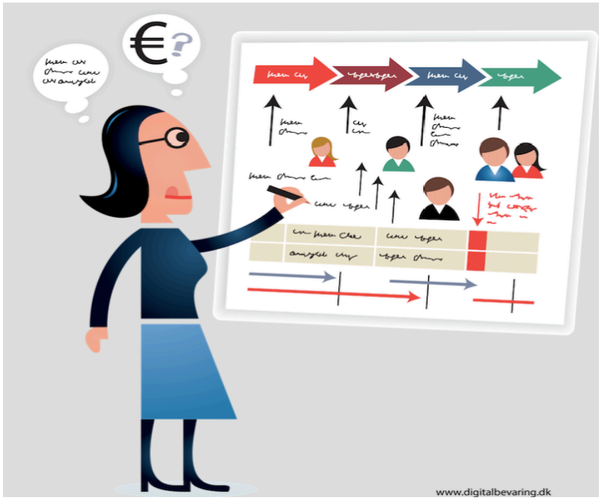
— Do you agree that drafting a DMP is useful?

A. Yes, it has already made me think about things I would otherwise forget
B. Yes, I think it can in theory be a useful exercise
C. No, I think it's just another administrative burden
D. I don't know

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GOOD PRACTICE FOR ANY PROJECT USING DATA

- Consider it a tool to help you
  - → increase awareness of RDM
  - → enhance data quality, sustainability & (re)usability
  - → comply with legal, ethical & other requirements
  - → identify any support & resources required
  - → clarify expectations & responsibilities
  - → starting point for a data paper?
- But of course: a plan cannot do the work for you!



'Planning' by Jørgen Stamp, attribution: digitalbevaring.dk, licensed under CC-BY 2.5 DK

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EXTERNAL FUNDER & UGENT REQUIREMENTS

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EXTERNAL FUNDER DMP REQUIREMENTS

	FWO	BELSPO	H2020*	ERC
Proposal	'DMP-light' (5 questions)	Provisional DMP	NO (RDM paragraph under 'Impact' section)	NO
Initial DMP	Month 6	Month 6	Month 6 (deliverable)	Month 6
Final DMP	YES (final report)	YES (final report)	YES (final report)	NO (but keep up to date)
Periodic report DMP	NO	NO	YES	NO
Funder template	YES (mandatory)	YES (recommended)	YES (recommended)	YES (recommended)

*DMP requirement also expected in Horizon Europe

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https://www.ugent.be/en/research/datamanagement/policies/external-funders.htm

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	RDM policy framework (2016)	BOF	IOF	PhD students**
DMP requirement?	NO (but encouraged for all researchers)	YES (for research granted from 2020 onwards*)	YES (for research granted from 2020 onwards)	YES (expected in OER for all new PhD students from 2020-2021)
Proposal	NA	NO	NO	TBC
Initial DMP	NA	Month 6	Month 6	TBC
Final DMP	NA	YES	YES	TBC
Template	NA	NO (any template in DMPonline.be)	NO (any template in DMPonline.be)	TBC

*Exceptions: BOF-ZAP positions, research equipment, coordination positions & collaborations

**Some Faculties already have a DMP requirement for PhD students.

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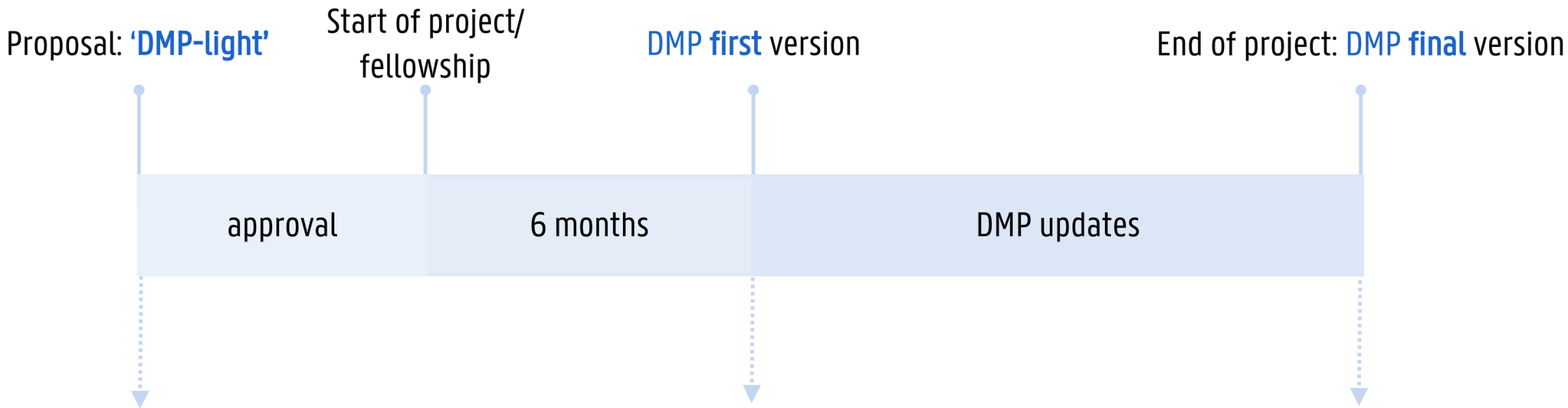
https://www.ugent.be/en/research/datamanagement/policies/ghent-university.htm

QUESTION TIME!

- If you are writing/submitting a DMP soon, who for?
  A. FWO
  B. Another external funder
  C. BOF or IOF
  D. My Faculty
  E. Myself
  F. Other

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PROCEDURE FWO-FUNDED RESEARCH



Answer 5 RDM questions in FWO application form.

Submit to Research Co-ordination Office no later than 6 months after official start date; FWO may request access.

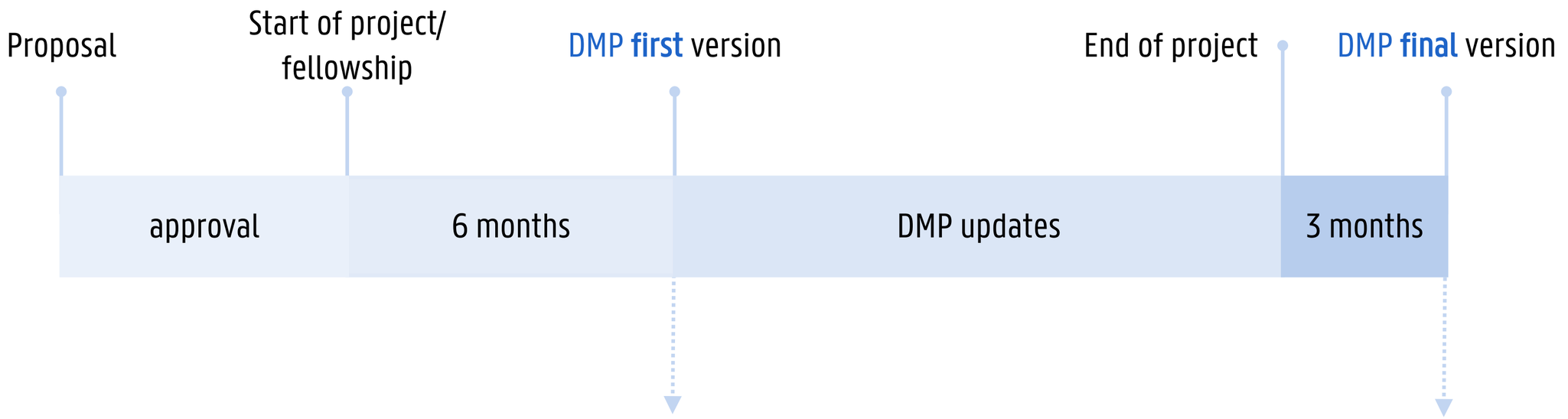
→ email PDF to vlaams-federaal@ugent.be

→ also submit copy to RCO if UGent is NOT main host institution

Submit to FWO along with report on completed research (part of final evaluation).

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PROCEDURE BOF & IOF-FUNDED RESEARCH



Submit no later than 6 months after official start date.

→ BOF: email PDF to bof@ugent.be

→ IOF: email PDF to iof@ugent.be

Submit no later than 3 months after end of project/fellowship.

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WHAT TO COVER IN YOUR DMP?

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VARIOUS TEMPLATES, BUT COMMON TOPICS

"Plans typically state what [meta]data will be created and how, and outline the plans for sharing and preservation, noting what is appropriate given the nature of the data and any restrictions that may need to be applied." (DCC website)



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FWO TEMPLATE: QUESTIONS ON 7 MAIN THEMES

Data description	
Ethical and legal issues	
Data documentation and metadata	webinar 19/05
Data storage and backup during the FWO project	webinar 2/06
Data preservation after the end of the FWO project	webinar 16/06
Data sharing and reuse	webinar 16/06
Responsibilities	

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DATA DESCRIPTION QUESTIONS

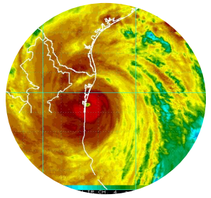
- Will you generate/collect new data and/or make use of existing data?
- Describe the origin, type and format of the data (per dataset) and its (estimated) volume.

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DATA DESCRIPTION – SOME TIPS

- Only provide answers relating to research data (not publications, presentations, etc.)
- Remember: ‘research data’ come in many different forms!
- Break down and list your data types conveniently: e.g. by technique, purpose, phase…
- Provide enough details for outsiders to understand the sort of data involved
  - → content, digital/non-digital, raw/processed…
  - → specify file formats (e.g. .csv, .txt, .docx…)
  - → specify data source/collection methods (e.g. experimental, observational, simulation… data)






ETHICAL & LEGAL ISSUES QUESTIONS





Will you use personal data?



Are there any ethical issues concerning the creation and/or use of the data (e.g. experiments on humans or animals, or dual use)?



Is there potential for tech transfer and valorization? Will IP restrictions be claimed for the data you created?



Do existing 3rd party agreements restrict dissemination or exploitation of the data you (re)use?

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ETHICAL & LEGAL ISSUES – SOME TIPS

Personal data

→ = any information relating to an identified or (in)directly identifiable living individual.

→ for example: *names, occupations, (email/IP) addresses, dates of birth, phone numbers, bank account numbers, physical characteristics, health-related information, voice recordings, photographs, political views, ethnicity, trade union membership, etc.*

→ processing is subject to General Data Protection Regulation

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https://www.ugent.be/en/research/datamanagement/policies/privacy.htm

QUESTION TIME!

- Does your research involve processing of personal data?
  A. Yes, and I know how to be GDPR-compliant
  B. Yes, but I am not sure how to comply with the GDPR
  C. No
  D. I don't know

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ETHICAL & LEGAL ISSUES – SOME TIPS

Ethical issues: 'yes' if

→ e.g. you need permission from ethical committee and/or informed consent for collecting & handling data (incl. archiving & sharing):

also see 'Ethics' section in FWO application form (research involving human participants, human embryos, human cells/tissues, personal data, animals, potential for military applications…)

→ e.g. you work with sensitive data (could cause harm if disclosed – e.g. to endangered species, vulnerable sites/groups, national security, public health…)

ETHICAL & LEGAL ISSUES – SOME TIPS

Valorization, IP & third-party agreements

→ restrictions on collecting, sharing, archiving & reusing data may be applicable

→ e.g. your research has commercial potential:
sharing may not be possible (to protect confidential knowhow), or only after embargo period (to seek patent protection first) – always contact UGent TechTransfer before disclosing anything!

→ e.g. you work with third-party data/collaborate with external partners:
check licenses/agreements to know what you can do

DATA DOCUMENTATION & METADATA QUESTIONS



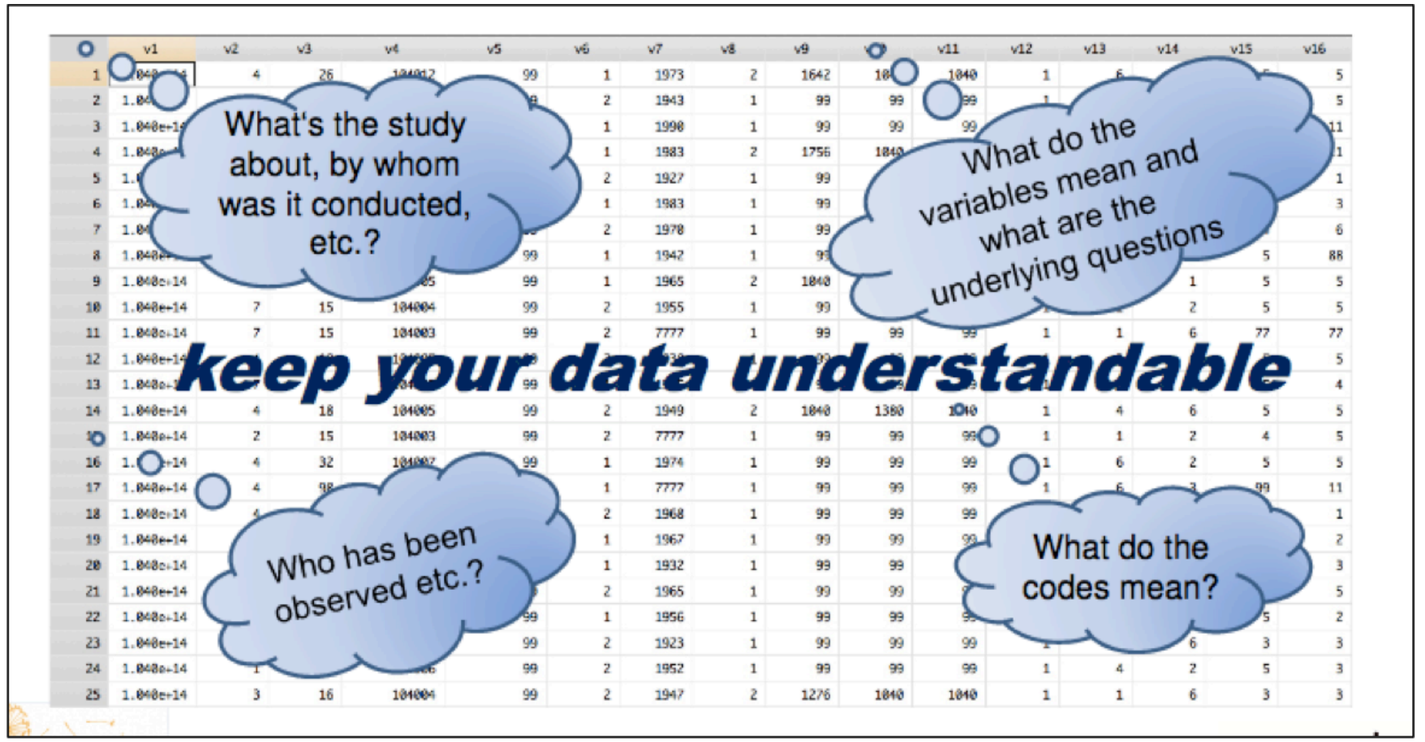
- What documentation will be provided to enable understanding and reuse of the data collected/generated in this project?
- Will a metadata standard be used?

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DATA DOCUMENTATION – SOME TIPS

Any information needed to understand & reuse research data, and to reproduce your research – ideally following standards in your domain

- → codebook/data dictionary explaining variables
- → dataset structure, relation between files
- → study context
- → protocols, methods, SOPs
- → notes/annotations
- → machine configurations
- → software used
- → processing & analysis scripts
- → consent information
- → etc.



From 'Data Handling: Documentation, Organization and Storage' by Sebastian Netscher, licensed under CC-BY

METADATA ('DATA ABOUT DATA') – SOME TIPS

A highly structured form of data documentation

→ needed to search & find data (e.g. catalogues)

→ also used to further describe/annotate data

→ fixed set of elements, as defined by metadata 'schema'

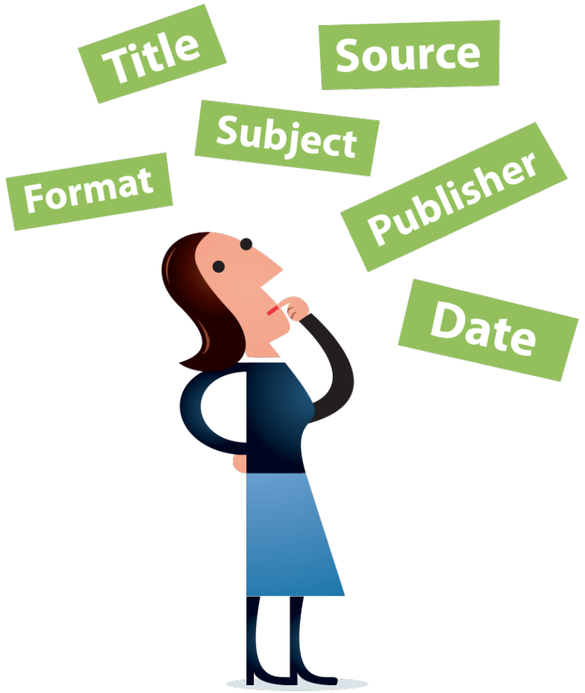
→ use existing standards!

- **generic**: e.g. Simple Dublin Core, Datacite
- **domain-specific**: check via

  fairsharing.org

  RDA metadata directory

  DCC's list of disciplinary metadata



Digitalbevaring.dk

'Metadata' by Jørgen Stamp, attribution: digitalbevaring.dk, licensed under CC-BY 2.5 DK

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https://www.ugent.be/en/research/datamanagement/during-research/documentation.htm

DATA STORAGE & BACKUP QUESTIONS

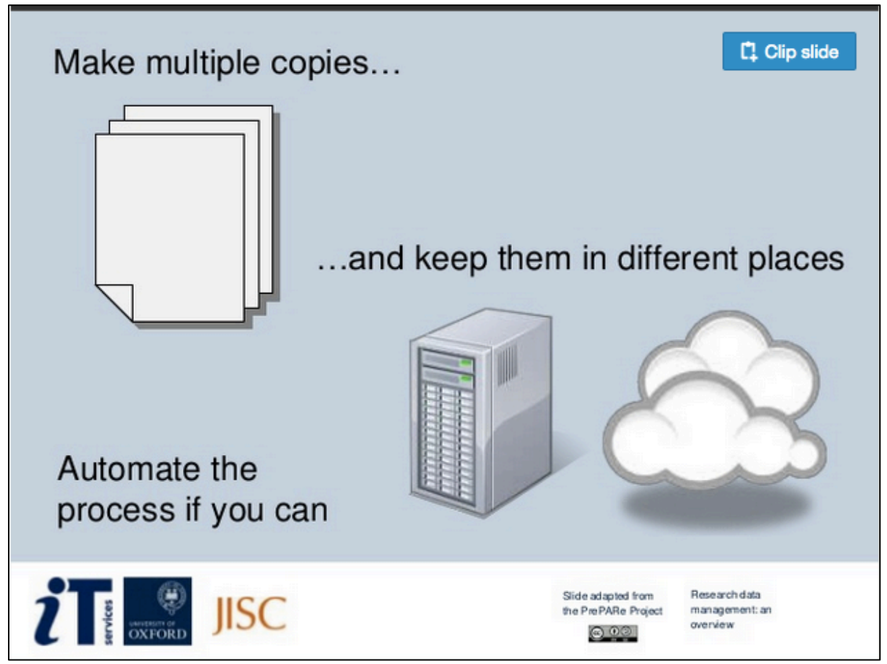


- Where will data be stored?
- How will the data be backed up?
- Is there sufficient storage & backup capacity during the project?
- What are the expected costs for data storage & backup during the project? How will these costs be covered?
- Data security: how will you ensure that the data are securely stored and not accessed or modified by unauthorised persons?

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DATA STORAGE & BACKUP – SOME TIPS

- Preferably use university-managed services, not (just) local storage devices.
- Storage capacity/costs
  - → indicate if you need additional storage beyond UGent provision
  - → if so: indicate what + how much this costs
- Security
  - → physical security
  - → security of computer systems & files
  - → network security
  - → check 10 tips for safely working with IT at UGent



From 'RDM: An Overview' by Research Support Team, IT Services (University of Oxford), licensed under CC-BY-NC-SA 4.0

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https://www.ugent.be/en/research/datamanagement/during-research/storage.htm
https://www.ugent.be/en/research/datamanagement/during-research/data-security.htm

DATA PRESERVATION QUESTIONS

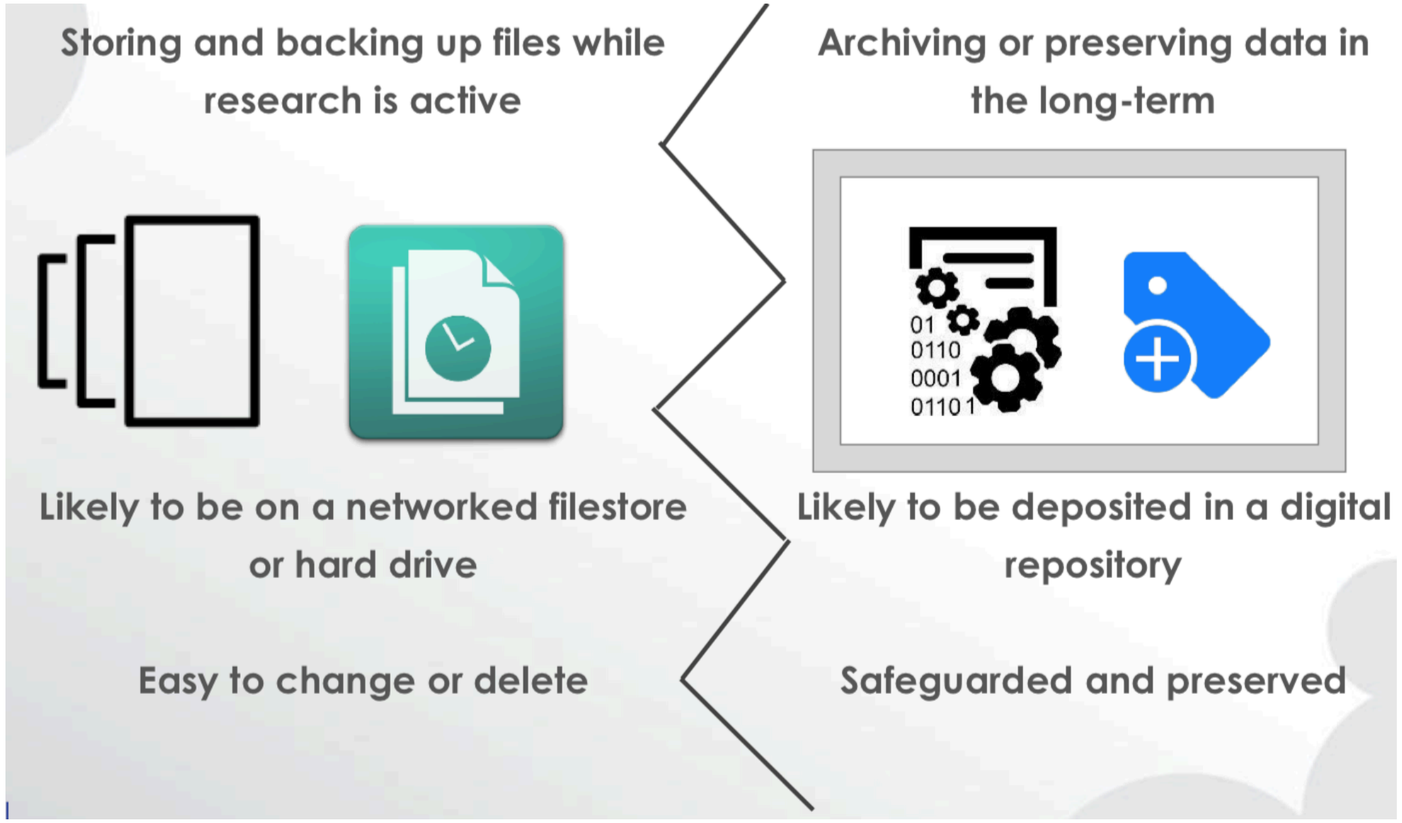


- Which data will be retained for the expected 5-year period after the end of the project?
- Where will these data be archived (= stored for the long term)?
- What are the expected costs for data preservation during these 5 years? How will the costs be covered?

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DATA PRESERVATION – SOME TIPS

Keep in mind the difference between storing & archiving data!



From 'What you need to know to prepare a data management plan' by Marjan Grootveld, licensed under CC-BY

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DATA PRESERVATION – SOME TIPS

- Deposit data in a trusted external data repository
  - → preferably **domain-specific** (if available)
  - → **general-purpose** (e.g. Zenodo, Dryad, OSF)
  - → find a suitable home via re3data.org
- But: some data may need to be kept in-house
- Costs
  - → e.g. for depositing data in a repository

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

seadatanet Search



https://www.fosteropenscience.eu/content/re3data-demo

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https://www.ugent.be/en/research/datamanagement/after-research/preservation.htm

DATA SHARING & REUSE QUESTIONS

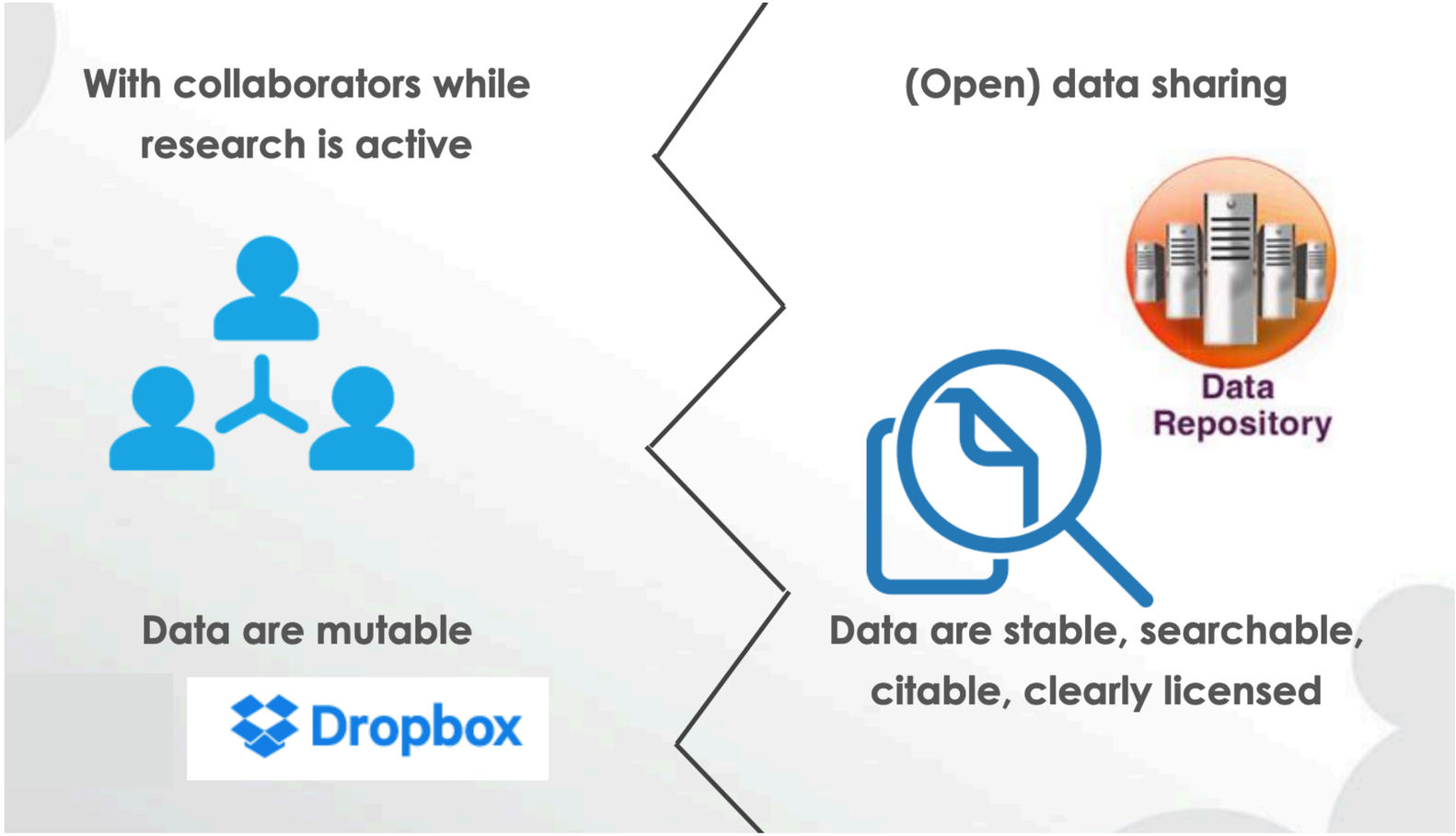


- Are there any factors restricting or preventing the sharing of (some of) the data?
- Which data will be made available after the end of the project?
- Where/how will the data be made available for reuse?
- When will the data be made available?
- Who will be able to access the data and under what conditions?
- What are the expected costs for data sharing? How will these costs be covered?

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DATA SHARING – SOME TIPS

What is meant?



From '[What you need to know to prepare a data management plan](#)' by Marjan Grootveld, licensed under [CC-BY](#)

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https://www.ugent.be/en/research/datamanagement/after-research/sharing.htm

DATA SHARING & REUSE – SOME TIPS

Data sharing is a spectrum!

"As open as possible, as closed as necessary"



Can be freely used, modified & shared by anyone for any purpose

http://opendefinition.org

open, standard licenses (e.g. CC0, CC-BY)

Limits on who can access & use data, how, or for what purpose, e.g.

→ only certain (types of) users
→ only certain types of use

managed access procedures, bespoke licenses/end user agreements

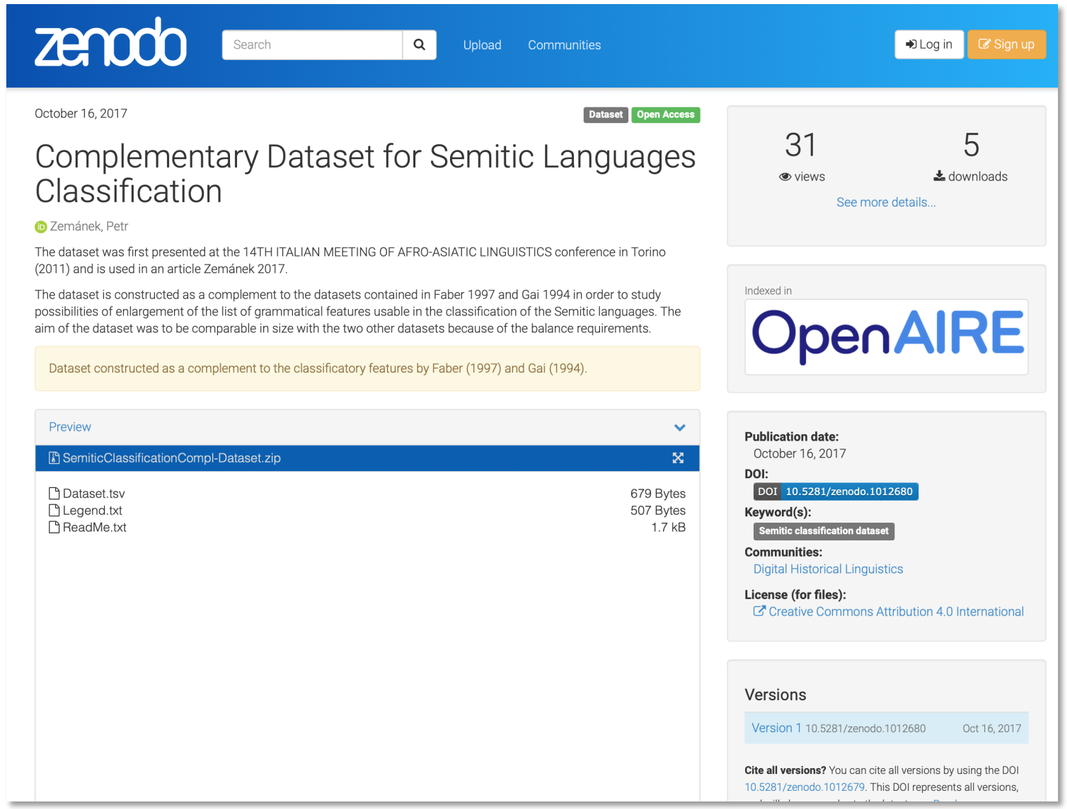
Under embargo
Unable to share

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Adapted from 'Managing and sharing research data' by S. Jones, CC-BY

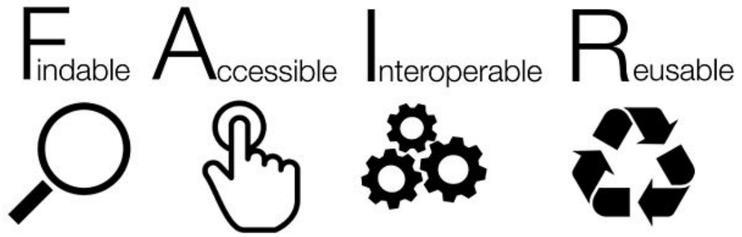
DATA SHARING & REUSE – SOME TIPS

- DO share via a data repository if possible.
- DON'T DIY/share "upon request by email".



https://doi.org/10.5281/zenodo.1012679

Data are findable, clearly licensed, citable (also see FAIR principles!).



By SangyaPundir - Own work, CC BY-SA 4.0

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RESPONSIBILITIES

QUESTIONS

- Who will be responsible for data documentation & metadata?
- Who will be responsible for data storage & backup during the project?
- Who will be responsible for ensuring data preservation & sharing?
- Who bears the end responsibility for updating & implementing this DMP?

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RESPONSIBILITIES – SOME TIPS

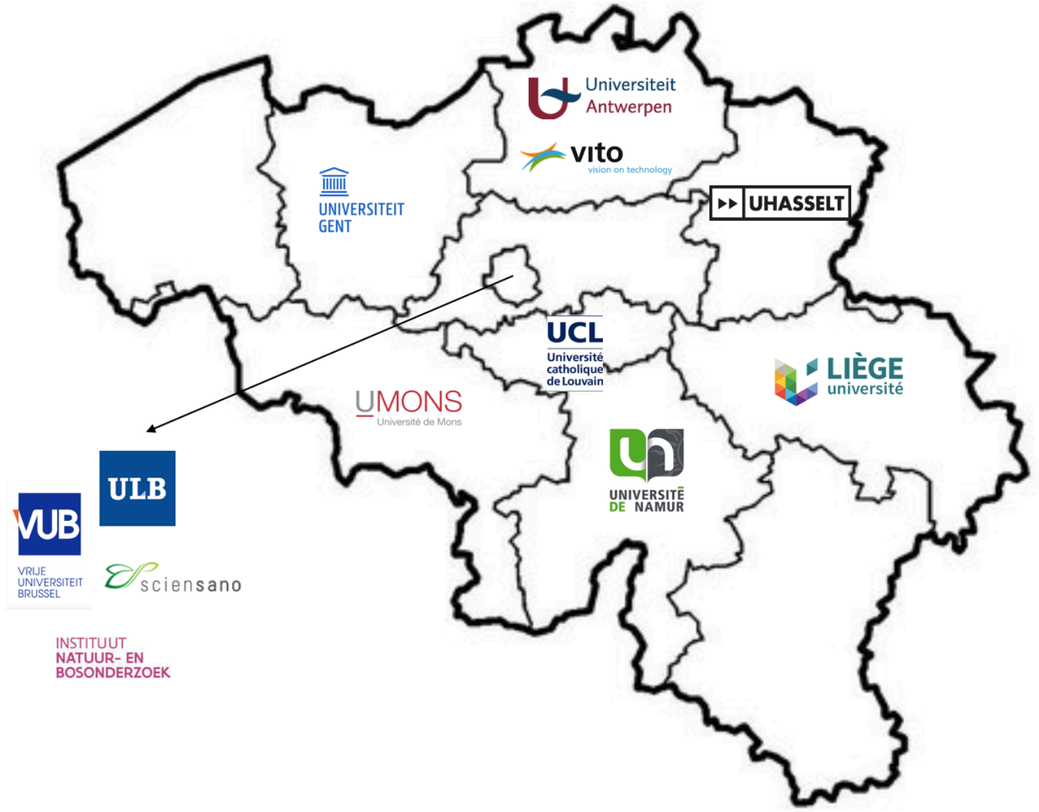
- By default, PI bears end responsibility for DMP & its implementation
- Responsibility for day-to-day data management may be
  - → shared among project team members, and with institutional services (e.g. IT service)
  - → outsourced to others (e.g. data repository, external partner)

DMPONLINE.BE

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DMPONLINE.BE

- Online planning tool
- Available to researchers from institutions in DMPbelgium consortium
- Contains DMP templates + guidance
  - → from funders (including FWO)
  - → or institutional



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https://dmponline.be

QUESTION TIME!

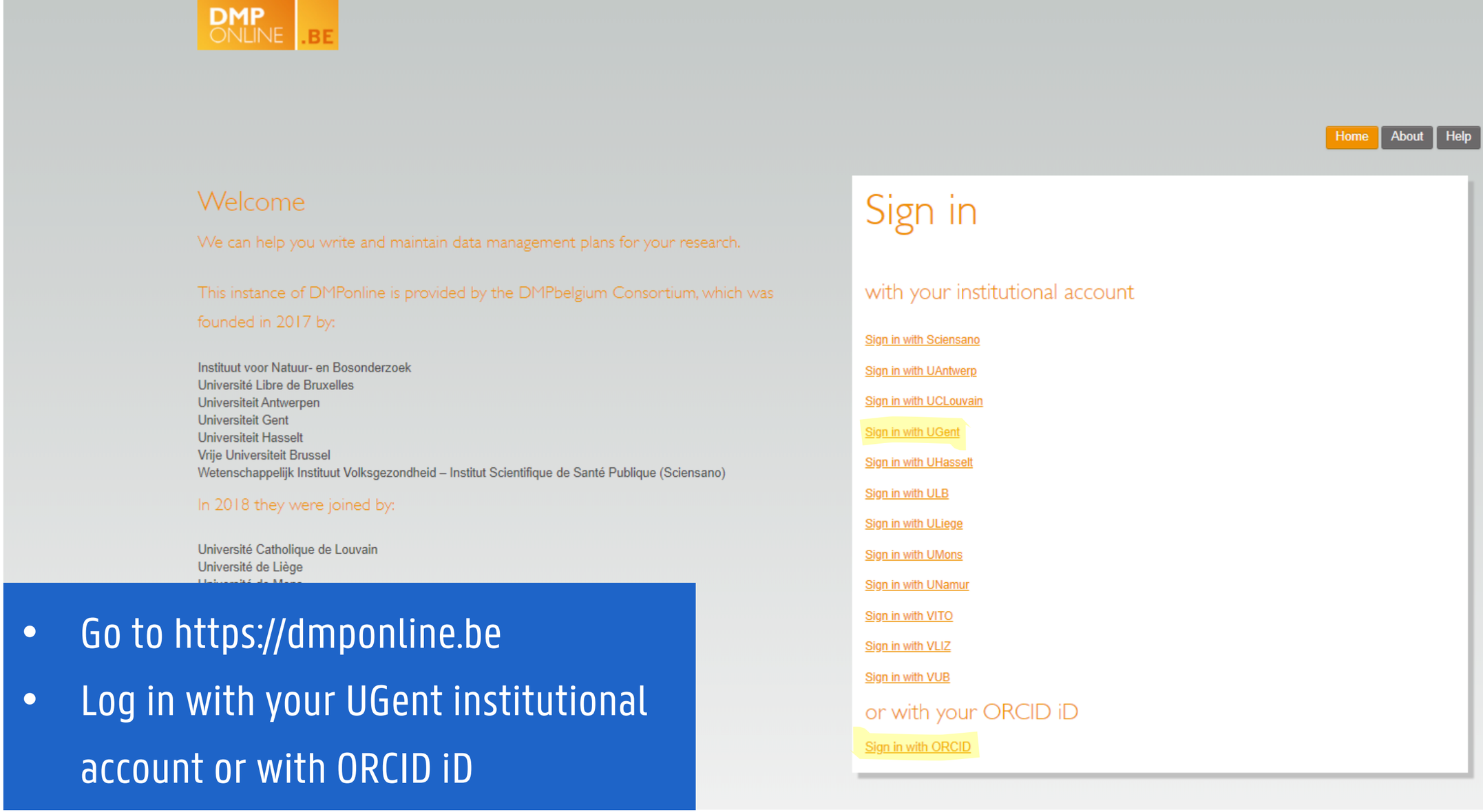
— Have you ever used DMPonline.be before?

A. Yes

B. No, but I didn't know about it

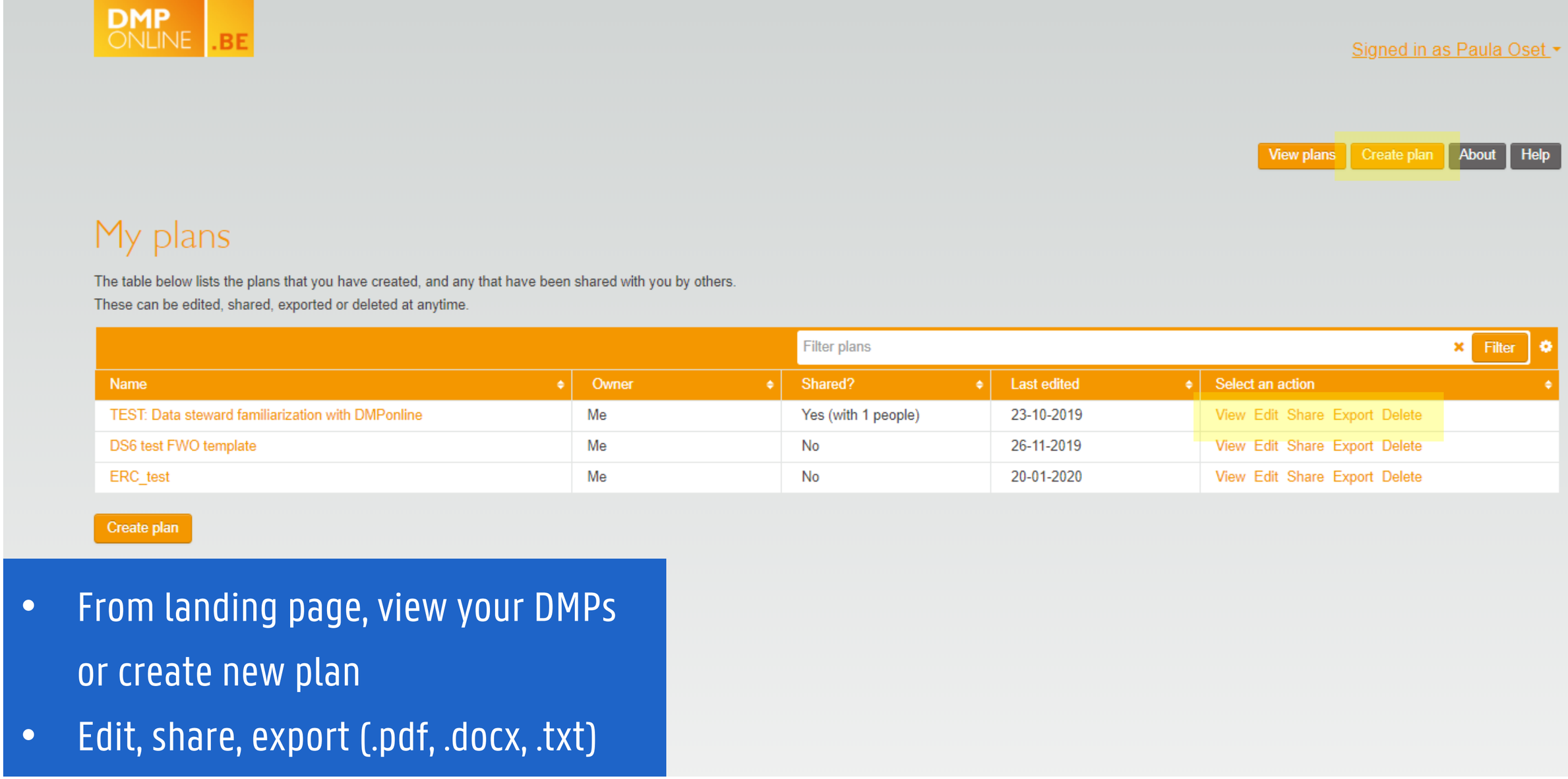
C. No, although I know about it

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- Go to https://dmponline.be
- Log in with your UGent institutional account or with ORCID iD

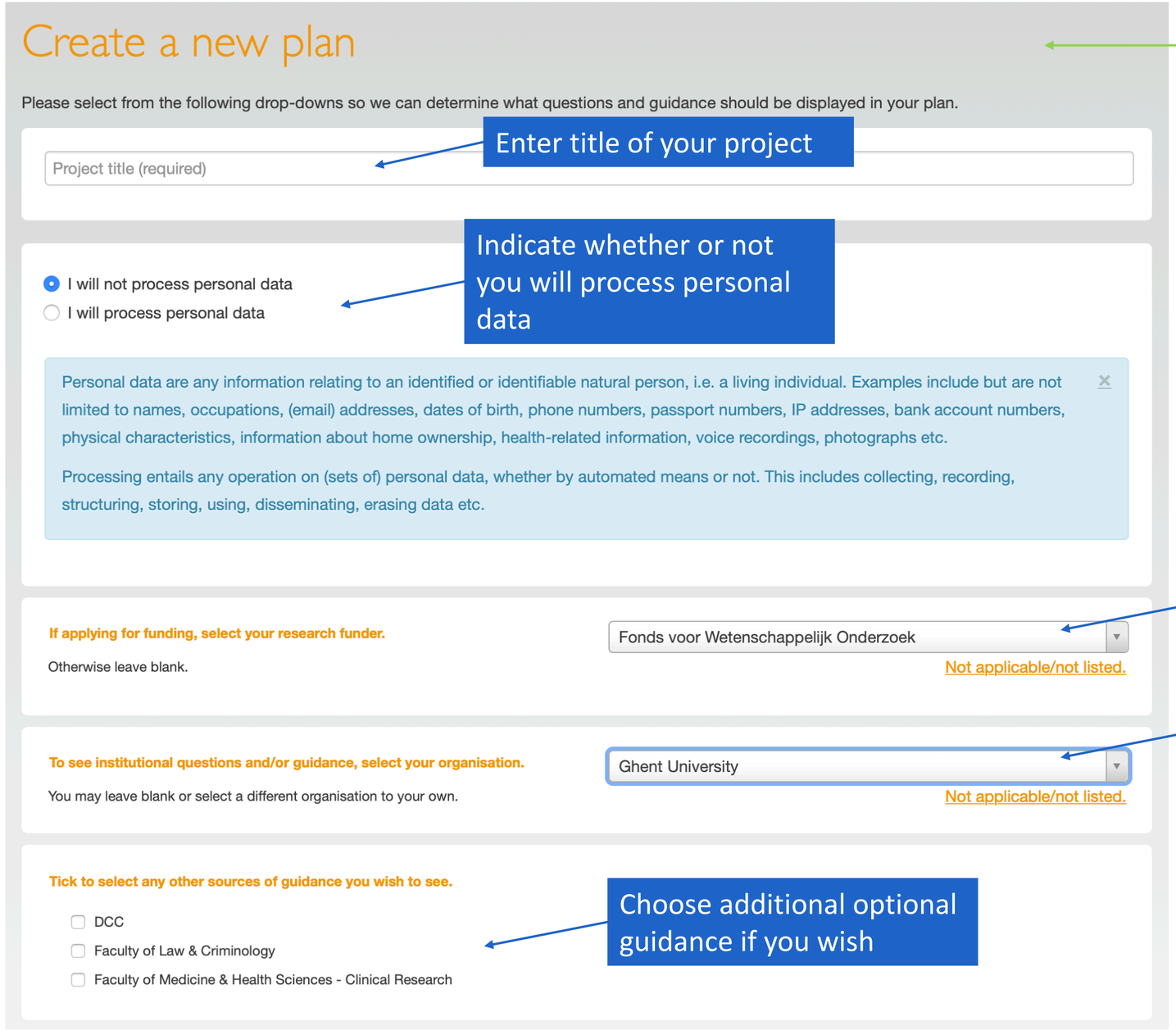
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- From landing page, view your DMPs or create new plan
- Edit, share, export (.pdf, .docx, .txt)

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‘CREATE A NEW PLAN’ WIZARD



Go through wizard to help tool determine which template + guidance to show

Enter title of your project

Indicate whether or not you will process personal data

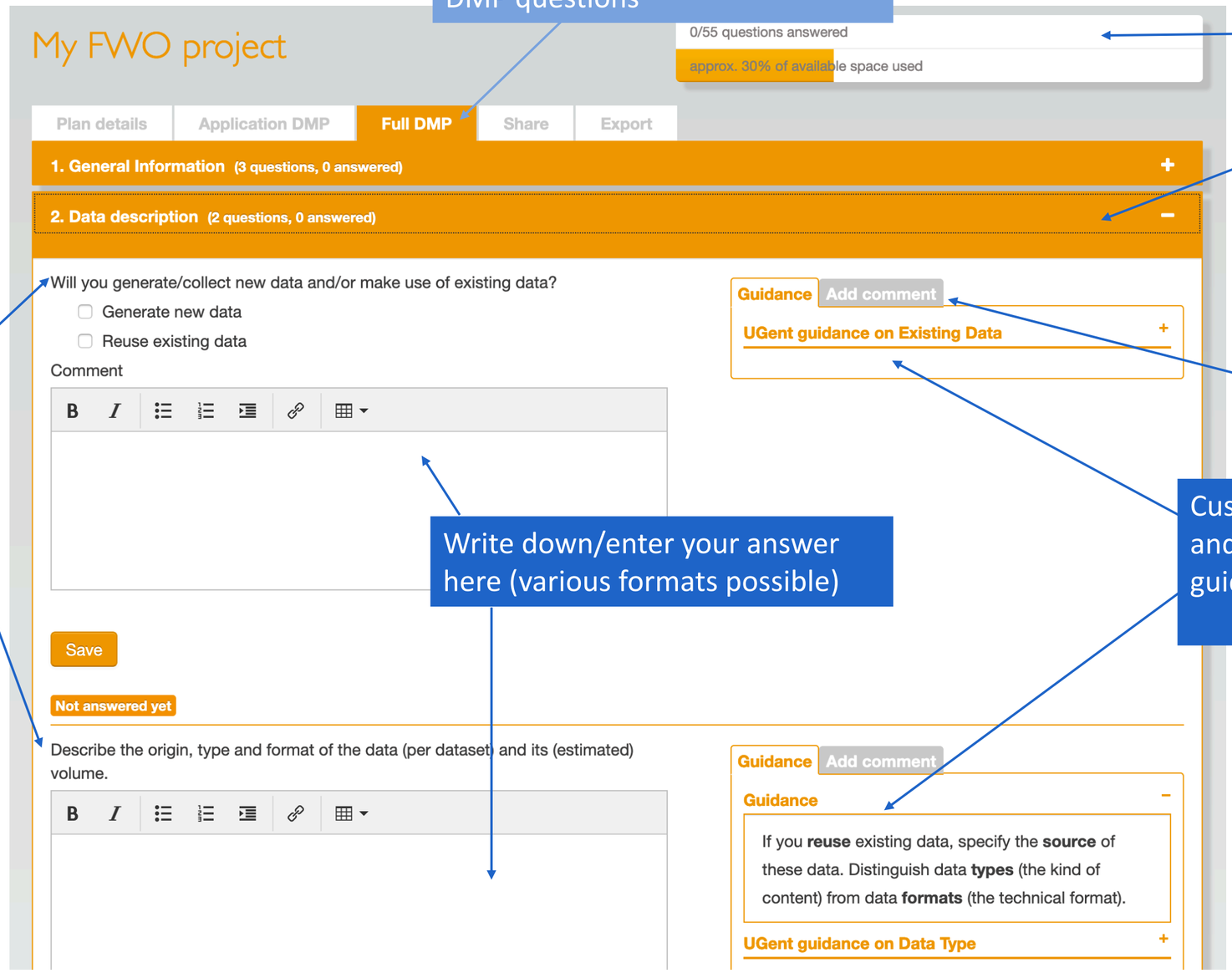
Select funder (e.g. FWO) to get its template

Select Ghent University to get local guidance, as well as institutional templates if funder not applicable

Choose additional optional guidance if you wish

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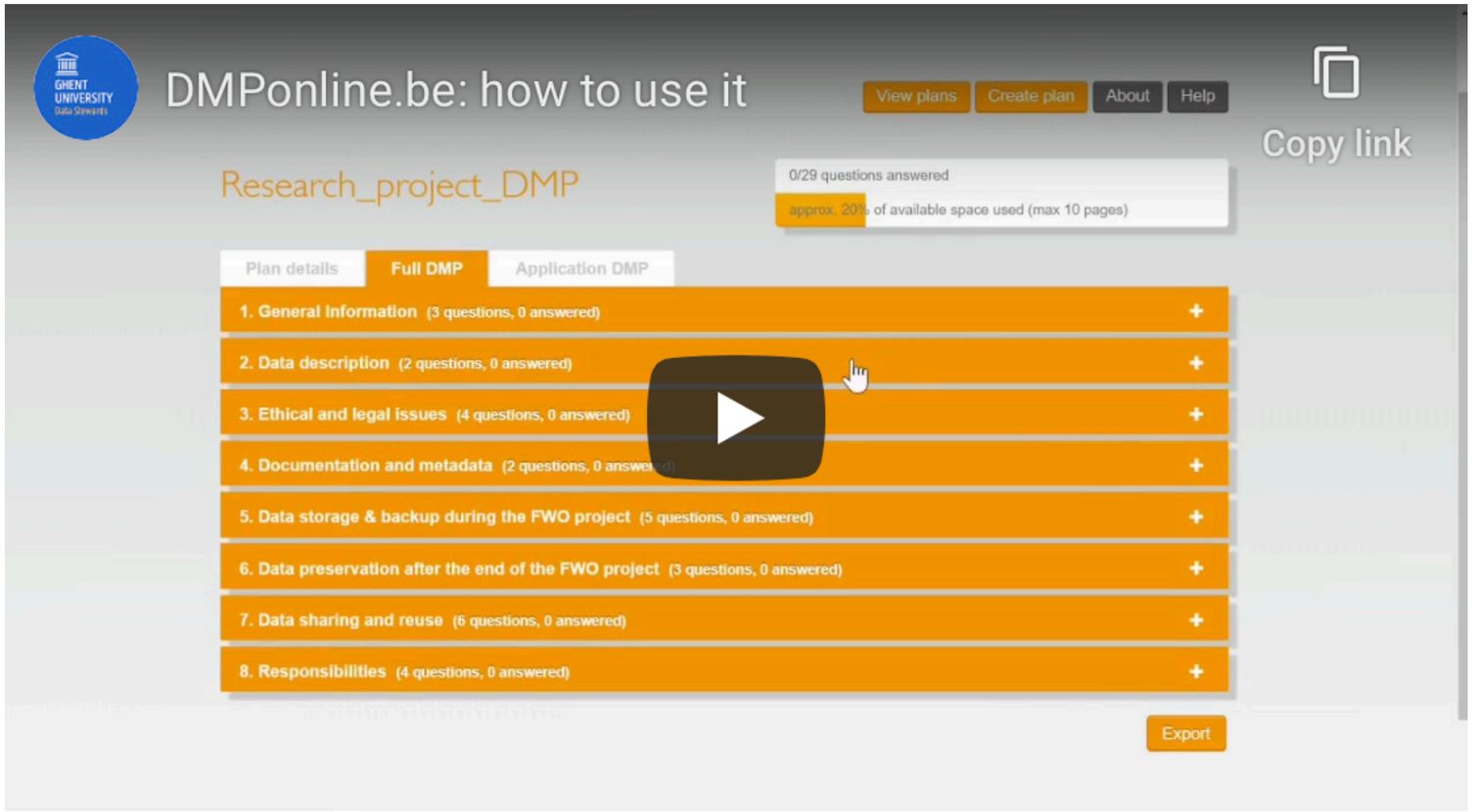
EDITING YOUR PLAN



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DMPONLINE.BE MANUAL

Available via https://researchtips.ugent.be/en/tips/00001281/



demo: https://youtu.be/kXuBC6Hl01g

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A NOTE ON WORKING WITH PERSONAL DATA

Does your (FWO) project involve research with personal data?

→ necessary to register personal data processing activities before data collection

→ meet this registration requirement at same time as drafting your DMP in DMPonline.be:

- tick **“I will process personal data”** option in ‘create plan’ wizard
- available DMP templates will now also include institutional **“GDPR” questions**
- select template of your choice
- complete mandatory GDPR questions – that’s it!
- when exporting DMP (e.g. for funder), you can leave out GDPR questions

→ more info on registration procedure: https://researchtips.ugent.be/en/tips/00001795/

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WHAT OTHER DMP SUPPORT IS AVAILABLE?

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EXAMPLE DMPS

- LIBER DMP catalogue (includes reviews)
- Examples on Digital Curation Centre website
- DMPs published in Research Ideas and Outcomes (RIO) OA journal

Data Management Plan Catalogue

Welcome to the Data Management Plan (DMP) Catalogue of the LIBER Research Data Management Working Group. This "DMP CAT" is an innovative approach to provide:

- A central hub for DMPs from different disciplines
- Quality reviews of the DMPs with assessments of the quality of the different parts

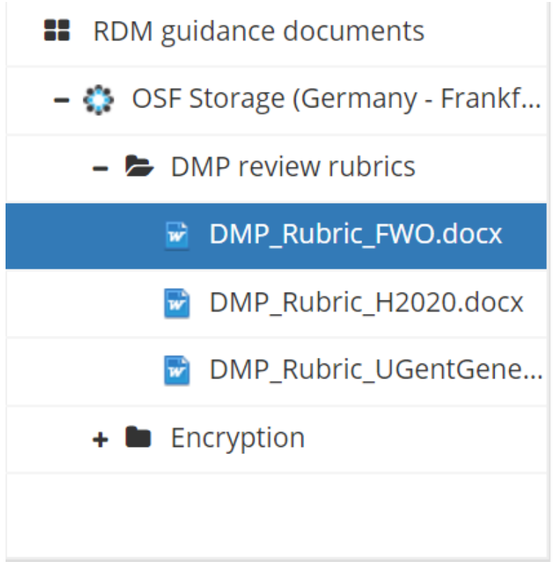
The purpose of this catalogue is to inspire researchers and others in the process of writing a Data Management Plan.

This DMP CAT provides the metadata and reviews in tabular form on this website and links to the citable and DOI-equipped PDF version of the DMPs on Zenodo. This means that the DMP are officially published via Zenodo: authors and contributors are properly listed and linked to their ORCID-ID, the DMP received as persistent link and is citable.

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DMP FEEDBACK

- Use an evaluation grid for self-assessment (FWO & other templates): https://osf.io/dhpr3/
- Request feedback from data steward team via rdm.support@ugent.be (allow enough time)



Version 2019-11-25

FWO DMP Evaluation Rubric

This is an evaluation rubric for the DMP template of the FWO (i.e. the FWO template for DMPs that have to be drafted for funded projects, not the RDM questionnaire that is part of the FWO application form). It is inspired by and based on the example rubric (CC0) from the DART project (https://osf.io/26b9r/), which was presented during the IDCC 2016 DMP Workshop in Amsterdam on 25 February 2016.

	PERFORMANCE LEVELS		
PERFORMANCE CRITERIA	**Complete/detailed**	**Addressed issue, but incomplete/too vague**	**Did not address issue**
1. Indicates whether the project will generate/collect **new data** and/or make use of **existing data**. **[Q2.1]**	Clearly indicates whether the project will generate new data and/or reuse existing data, and explains why. E.g.: *"The project will reuse existing datasets on local mobility in the city of Ghent. These data will provide sufficient information to perform a quantitative analysis of the use of public transport services. However, to allow a qualitative analysis of local perceptions of public transport, the project will also generate new data, as there are no existing qualitative data about that aspect of our study."* E.g.: *"The geological data generated in the 'Ancient*	Indicates whether the project will generate new data and/or reuse existing data, but does not or only vaguely explain why.	There is no indication as to whether newly generated and/or existing data will be used.

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https://www.ugent.be/en/research/datamanagement/before-research/datamanagementplan.htm

TOOLS AND GUIDANCE TO BUDGET FOR RDM

https://www.ugent.be/en/research/datamanagement/before-research/costs.htm

3. Data Storage & Back-up	**Data backup** Does the institution provide regular backup or not? Consider how frequently backups should be done, how many backups should be stored.	Institutional backup – included in standard indirect cost/overheadsadditional backup needed – cost according to number of copies to be kept, frequency of backup and storage media needed	Examples: University drive €0.80 per GB/y Cloud: €0.30 per GB/y2 x Harddrive: €0.14 per GB (single purchase)
3. Data Storage & Back-up	**Data storage** How much data storage space is needed for the entire duration of the project? Do you need to set up a data model and	If storage is provided by the institution – cost is included in standard indirect costs or overheads If additional storage needed – cost server/ disk space, as well as the cost of setting up and maintenance	Example: Cloud Database as a service: €160/Month (storage 5GB transfer 30GB)Database architect at level 2* salary

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WRAP-UP

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DON'T BE SCARED OF DMP WRITING!

Some final tips:

→ keep it simple, but be as specific as possible.

→ if you don't know yet, just say so (& update your plan later).

→ not necessarily right/wrong answers, but justify decisions.

→ familiarize yourself with RDM terminology & best practices (e.g. via UGent or external training)

→ also see it as an opportunity to inform us about your RDM needs.



TOP 10 excuses for not having a DMP by @FranziMachtDas, CC-BY-SA 4.0

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Q&A

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Q&A

Tu 19 May	**13:00**	**Organizing & documenting data**
Tu 02 Jun	13:00	Storing data & data security
Tu 16 Jun	13:00	Preserving & sharing data

ugent.be/en/research/datamanagement

osf.io/57gvr/

@UGentRDM

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UGent Data Stewards

RESEARCH DEPARTMENT - UNIVERSITY LIBRARY

rdm.support@ugent.be

@UGentRDM

ugent.be/en/research/datamanagement

www.ugent.be

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